Panga na Kupangisha – Real Estate & Rental Investment System

SRS / Project Proposal

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1. Introduction

1.1 Purpose

The purpose of this system is to create a comprehensive, Tanzanian-compliant digital platform that connects landlords, tenants, investors, and service providers. The system will: - Facilitate long-term rentals (monthly leases). - Provide investment opportunities in rental properties. - Enable renters to request financial support (Rent Assistance / Microloan). - Ensure compliance with Tanzanian legal and regulatory frameworks (NaPA, TIN, Land Registry, TALA). - Include advanced features like analytics, predictive insights, virtual tours, and secure communication.

This system is targeted for mobile-first access, with backend API support via Django, real-time features (chat, notifications), and optional AR/VR tours.

1.2 Scope

The system will cover: - Role-based access: Landlords, Tenants/Renters, Investors, Property Managers, Vendors/Contractors, Admin/Super Admin. - Multi-unit property management with complete property data, images, virtual tours, and location validation via NaPA. - Lease and rental application management including background screening and digital lease signing. - Payment processing, investment tracking, and microloan/rent-assistance management. - Maintenance requests, work order assignment, and automated notifications. - Analytics dashboards for landlords, investors, tenants, and administrators. - Legal compliance for landlord listings, including tax, TIN, property licenses, and title verification.

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2. Stakeholders

- **Landlords:** List properties, track tenants, manage rent, access investment features.
- **Tenants/Renters:** Browse listings, apply for rentals, pay rent, submit maintenance requests, request financial assistance.
- **Investors:** Invest in properties, track ROI, receive payouts.
- **Property Managers:** Manage multiple properties, oversee maintenance, approve tenants.
- **Vendors/Contractors:** Accept maintenance tasks, submit invoices.
- **Admin / Super Admin:** System oversight, dispute resolution, analytics, compliance monitoring.
- **CEO / Company Stakeholders:** Oversee company compliance, revenue, and platform expansion.

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3. Functional Requirements

3.1 Authentication & Roles

- Phone number login + OTP.
- Role-based access control (RBAC).
- Profile management for all user types.

3.2 Property Management (Landlord Module)

Essential Attributes for Compliance & Visibility: - Landlord ID verification (NIDA/Passport, corporate documents if applicable). - Proof of property ownership (Certificate of Title / Occupancy). - NaPA-compliant address (street, house number, ward, district, GPS coordinates). - Property details: Type, floor area, rooms, bathrooms, amenities, construction year, photos, virtual tours (optional AR). - Compliance: Building plan approval, occupancy permit, land tenure verification. - Licensing: Tourism license (TALA) if short-term rental, estate agent license if using a broker, business permits if multiple units. - Tax compliance: TIN, VAT (if applicable), property tax certificates. - Security: Document verification, sensitive data encryption, address validation via NaPA, transparency of ownership.

3.3 Rental Applications & Lease Management

- Online applications with document uploads.
- Tenant background, credit, and eviction checks.
- Digital lease agreements with e-signatures.
- Lease renewal, termination, and prorated calculations.

3.4 Payments & Financial Management

- Recurring rent and autopay via MPESA, cards, bank transfer.
- Investment dashboard for investors.
- Escrow system for rent assistance/microloan funds.
- Accounting and reporting: Ledgers, income statements, tax reminders.

3.5 Rent Assistance / Microloan Feature

- Tenants post rent-assistance requests (gift or microloan).
- Public or private request visibility.
- Track contributions, remaining balance, and funding deadlines.
- Optional repayment schedules with automated reminders.

3.6 Maintenance & Services

- Tenants submit maintenance requests.
- Landlords/property managers assign vendors.
- Track progress, completion, and payments.
- Preventive maintenance schedules and inspection logs.

3.7 Communication

- In-app chat between users (tenant-landlord, investor-landlord, tenant-vendor).
- Notifications: Rent due, maintenance updates, funding received.
- Automated messaging templates.

3.8 Analytics & Dashboards

- Landlord: Occupancy, rent collection, revenue per property.
- Investor: Portfolio performance, ROI, payouts.
- Tenant: Lease status, payment history, support requests.
- Admin: Platform usage, compliance reports, KPIs.
- Predictive insights: Rent pricing suggestions, investment opportunities.

3.9 Reviews & Ratings

- Tenant reviews for property and landlord.
- Landlord reviews for tenant.
- Investor feedback system.

3.10 Integrations

- Google Maps / NaPA APIs for addresses and locations.
- Payment gateways (MPESA, Stripe, mobile money).
- Tenant screening services.
- Email/SMS services (Twilio, SendGrid).
- Optional IoT / smart home integrations.

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4. Non-Functional Requirements

- **Scalability:** Support thousands of users and properties.
- **Performance:** Fast search (<2 seconds), dashboard loading optimized.
- **Reliability:** 99.9% uptime, backups, disaster recovery.
- **Security:** Data encryption, RBAC, PCI compliance, PII protection.
- **Usability:** Mobile-first, responsive UI, accessibility compliant.
- **Maintainability:** Modular code, well-documented, CI/CD-ready.

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5. Proposed Database Modules

- Users: Landlord, Renter, Investor, Property Manager, Vendor, Admin.
- Properties: Units, NaPA-compliant addresses, amenities, licenses, documents.
- Rental Applications & Leases.
- Payments & Transactions: Rent, investments, microloan contributions.
- Maintenance & Work Orders.
- Messaging & Notifications.

- Rent Assistance Requests & Contributions.
- Reviews & Ratings.
- Analytics & Reports.

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6. Compliance & Legal Considerations

- Landlord property listings verified against Tanzania Land Registry & ILMIS.
- Tax obligations (TIN, VAT if applicable, property tax) must be provided by landlords.
- Addresses validated via NaPA.
- Sensitive documents encrypted; public visibility only shows necessary info.
- Platform and CEO/stakeholders comply with Tanzanian company registration and tax laws.

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7. Deliverables

1. Full SRS / Project Proposal PDF.
2. Detailed Database Model (ER diagram & tables).
3. Initial App Module Blueprint (mobile + backend APIs).
4. Compliance checklist for landlords to ensure valid property listings.

Summary: This system is a full-scale, compliant, and highly scalable platform for real estate and rental investments in Tanzania. By integrating rent assistance, investment opportunities, maintenance, analytics, and legal compliance, it is positioned as the most professional and essential property management ecosystem in the market.